Find the linear classifier for an OR gate using Logistic Regression

The linear equation for a two input OR gate is **w.x** = w1x1+w2x2+b. Let us assume w1=0.5, w2=0.75 and b = -1.25. So the initial z-score is

z = 0.5x1+0.75x2-1.25

1^{st} Cycle			
x1=0, x2=0, y=0	**w.x+b=-1.25**	ỹ = σ(w.x+b) = 0.22 h**w**(**x**)= 0	y ==hw(x) w1,w2,b remain same
x1=0, x2=1, y=1	**w.x+b=-0.50**	ỹ = σ (w.x+b) = 0.38 h**w**(**x**)=0	y /= hw(x) w1,w2,b need to updated
		w1 = .5+.2(1-0.38)(0)=0.5 w2=.75+.2(1-0.38)(1)=0.87 b=-1.25+.2(1-0.38)=-1.13	
		z=0.5x1+0.87x2-1.13	
x1=1, x2=0, y=1	**w.x+b=-0.63**	ỹ = σ (w.x+b) = 0.35 h**w**(**x**)=0	y /= hw(x) w1,w2,b need to be updated
		w1 = .5+.2(1-0.35)(1)=0.63 w2=.87+.2(1-0.35)(0)=0.87 b=-1.13+.2(1-0.35)=-1.0	
		z=0.63x1+0.87x2-1.0	
x1=1, x2=1, y=1	**w.x+b=0.5**	ỹ = σ (w.x+b) = 0.62 h**w**(**x**)=1	y == hw(x) w1,w2,b remain same
		w1=0.63 w2 = 0.87 b = -1.0	
		z=0.63x1+0.87x2-1.0	
2^{nd} Cycle			

x1=0, x2=0, y=0	**w.x+b=-1.0**	ỹ = Ϭ (w.x+b) = 0.27 h**w**(**x**)=0	y ==hw(x) w1,w2,b remain same
x1=0, x2=1, y=1	**w.x+b=-0.13**	ỹ = Ϭ(w.x+b) = 0.47 h**w**(**x**)=0	y /= hw(x) w1,w2,b need to be updated
		w1 = .63+.2(1-0.47)(0)=0.63 w2=.87+.2(1-0.47)(1)=1.78 b=-1.0+.2(1-0.47) = -0.09	
		z=0.63x1+1.78x2-0.09	
x1=1, x2=0, y=1	**w.x+b=0.54**	ỹ = Ϭ (w.x+b) = 0.63 h**w**(**x**)=1	y = hw(x) w1,w2,b remain same
		z=0.63x1+1.78x2-0.09	
x1=1, x2=1, y=1	**w.x+b=2.32**	ỹ = Ϭ (w.x+b) = 0.91 h**w**(**x**)=1	y == hw(x) w1,w2,b remain same
		Z = 0.63x1+1.78x2-0.09	